F70LA-S1 – Life Insurance Mathematics A, 2025/26
Chapter 1 – Preliminaries

1.1 Probability reminder

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space on which all the random variables in this course are defined.

1.1.1 Probabilities

Let A and B be two events, i.e. $A, B \in \mathcal{F}$. Then

$$\mathbb{P}[A \mid B] = \frac{\mathbb{P}[A \cup B]}{\mathbb{P}[B]}.$$

$$\mathbb{P}[A \text{ and } B] = \mathbb{P}[A \cap B] = \mathbb{P}[A \mid B]\,\mathbb{P}[B].$$

$$\mathbb{P}[A \text{ or } B] = \mathbb{P}[A \cup B] = \mathbb{P}[A] + \mathbb{P}[B] - \mathbb{P}[A \cap B].$$

If A and B are disjoint, i.e. $A \cap B = \emptyset$, then $\mathbb{P}[A \cap B] = \mathbb{P}[\emptyset] = 0$ and so $\mathbb{P}[A \text{ or } B] = \mathbb{P}[A \cup B] = \mathbb{P}[A] + \mathbb{P}[B]$.

1.1.2 Random variables

A random variable is a measurable function $X : \Omega \to \mathbb{R}$ from the sample space Ω to the set of possible outcomes $\mathbb{R}$. Measurability is a technical condition that relates a sigma-algebra generated by the open subsets of $\mathbb{R}$ to the sigma-algebra $\mathcal{F}$ via the function X. We don't need to worry about measurability in this course.

The **cumulative distribution function (CDF)** of X is the function F_X

$$F_X(x) = \mathbb{P}\left[X \leq x\right], \quad \text{for } x \in \mathbb{R}.$$

The probability that X takes values in the range $(a, b]$, where $a < b$, is

$$\mathbb{P}\left[a < X \leq b\right] = F_X(b) - F_X(a).$$

1.1.3 Discrete–valued random variables

If X is a discrete–valued random variable then

$$X = x_j \qquad \text{with probability } p_j \text{ for } j = 1, 2, \ldots,$$

with, for $j = 1, 2, \ldots$, $x_j \in \mathbb{R}$ and $p_j \in [0, 1]$ and $\sum_{k=1}^{\infty} p_k = 1$.

The CDF of a discrete–valued random variable will have jumps at the values $x_1, x_2, \ldots$, i.e. its CDF is a discontinuous function.

The **probability mass function (PMF)** of X is a function $p_X : \mathbb{R} \to [0, 1]$ such that

$$p_X(x) := \mathbb{P}[X = x], \quad \text{for } x \in \mathbb{R}.$$

It should be clear that $p_X(x_j) = p_j$, for $j = 1, 2, \ldots$, and otherwise $p_X(x)$ takes value zero.

The **mean** or **expected value** of X is

$$\mathbb{E}(X) := \sum_{j=1}^{\infty} x_j p_j.$$

The **variance** of X is

$$\mathbb{V}ar(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \sum_{j=1}^{\infty} x_j^2 \, p_j - (\mathbb{E}(X))^2 = \sum_{j=1}^{\infty} p_j \, (x_j - \mathbb{E}(X))^2 .$$

The **standard deviation** of X is

$$\mathbb{SD}(X) := \sqrt{\mathbb{V}ar(X)}.$$

1.1.4 Continuous–valued random variables

If Y is a continuous–valued valued random variable then its CDF is continuous everywhere.

The probability density function (pdf) of Y is a function

$$f_Y(y) := \frac{dF_Y(y)}{dy}, \quad \text{for } y \in \mathbb{R}.$$

Note that F_Y is a well-defined function as the CDF F_Y is continuous everywhere.

Also note that for very small values of $\Delta t > 0$, $f_Y(t)\Delta t \approx \mathbb{P}[t < Y \leq t + \Delta t]$.

The **mean** or **expected value** of Y is

$$\mathbb{E}(Y) := \int_{-\infty}^{\infty} t f_Y(t) dt.$$

The **variance** of Y is

$$\mathbb{V}ar(Y) := \mathbb{E}(Y^2) - (\mathbb{E}(Y))^2 = \int_{-\infty}^{\infty} t^2 f_Y(t)dt - (\mathbb{E}(Y))^2 .$$

The **standard deviation** of Y is

$$\mathbb{SD}(Y) := \sqrt{\mathbb{V}ar(Y)}.$$

1.2 Calculus reminder

Integration by parts. For continuously-differentiable functions $f, g : \mathbb{R} \to \mathbb{R}$,

$$\int_a^b f(x) \left(\frac{d}{dx} g(x)\right) dx = [f(x)g(x)]_a^b - \int_a^b \left(\frac{d}{dx} f(x)\right) g(x)dx.$$

1.3 Basic life insurance policies

Read Dickson, Hardy & Waters, Chapter 1.

- A life insurance (assurance) policy is a contract between the policyholder (usually the life insured) and the life office.
- The policyholder has to pay money to the life office for the contract. This can either be a single payment (called a single premium) or a regular series of payments (called regular premiums). The time period over which premiums are due to be paid is called the *premium-paying term*.
- Premiums stop if the policyholder dies before the premium-paying term ends.
- In return, the life office either will or may pay out money (whether it is certain or contingent on an event - like the policyholder dying - depends on the type of policy)
- The life office pays a lump sum (called either the sum assured or sum insured) if/when a specified event occurs during the *policy term*.
- The specified event is usually death within the term or survival to the end of the term.
- The premium-paying term may be the same as the policy term. If not, it is shorter.

- The premiums are invested by the life office and the proceeds are used to pay for the sum insured and the life office's expenses.

1.3.1 Basic types of life insurance policies

These are policies which pay out either upon death or survival of the insured life or lives.

- Whole life policy: Sum insured paid on the death of the policyholder.
- Term insurance: Fixed policy term; sum insured paid on the death of the policyholder if that occurs within the policy term.
- Endowment insurance policy: Fixed policy term; sum insured paid on the death of the policyholder if that occurs within the policy term or at the end of the policy term if the policyholder is still alive.
- Pure endowment policy: Fixed policy term; sum insured paid at the end of the policy term if the policyholder is still alive.

1.3.2 Life annuities

These are policies which pay out as long as the insured life or lives are alive.

- Benefit is a series of regular payments starting at a specified time and continuing for a specified time (which could be unlimited). Payments are made only if the policyholder is still alive at the time the payment is due.
- A life annuity is a common form of benefit for a pension policy.